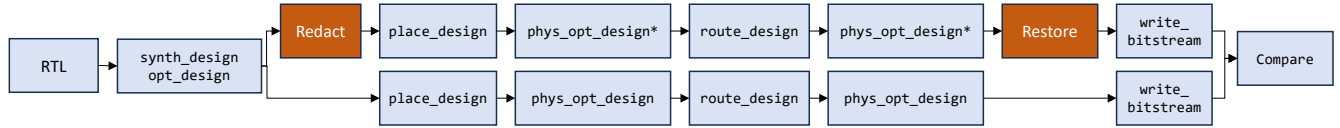


CAD-in-the-Cloud: Protecting FPGA Design Privacy via Redacted Netlists

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ABSTRACT

We propose a design flow to hide FPGA design functionality by redacting key properties during cloud compilation. The original design functionality can be restored post-implementation, with no change to the produced bitstream.

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Overview

As FPGAs grow in complexity, CAD tools require increasing computational resources to obtain high-quality results. Cloud computing offers on-demand high-performance resources at lower cost than local hardware. This enables parallelized design space exploration (DSE) for timing closure, eliminates setup barriers for new users and educational environments, and allows vendors to provide seamless tool updates. However, designers are reluctant to expose proprietary netlists to cloud providers who could reverse engineer their IP. While secure servers protect designs from other customers, they do not protect against the cloud provider itself.

Approach

We propose a redaction-and-restoration flow that removes cell properties from post-synthesis netlists before sending them to a cloud provider for implementation. As placement and routing algorithms rely primarily on structural connectivity, not logic functionality, we can redact properties like LUT INIT values, BRAM contents, and DSP configurations while preserving the netlist structure. This allows us to offload the computationally expensive implementation phase while keeping design details confidential.

The flow consists of four steps: (1) Local synthesis and logical optimization, (2) Redaction of cell properties with original values saved to metadata, (3) Remote implementation (placement, routing,

physical optimization), and (4) Local restoration of original properties. The resulting bitstream can be identical to one generated without redaction.

Implementation

We implemented a proof-of-concept using Xilinx Vivado and RapidWright, targeting an Artix-7 200T FPGA. During redaction, LUT INIT strings are replaced with XOR functions of their inputs, and other properties are replaced with zeros or default values. After cloud implementation, each cell is matched by name and original properties are restored. We focus on the implementation phase because it consumes roughly 57% of total CAD runtime, and must be re-run on the entire design even for incremental changes, making it the dominant cost for design iterations.

Results

We tested on 30 benchmarks from VTR and OpenCores, ranging from 176 to 65,751 cells. Most cell properties can be safely redacted: for LUTs, flip-flops, and most other cells, all properties can be redacted. For DSPs and BRAMs, some configuration properties (port widths, register enables, operating modes) must remain visible to avoid DRC violations and allow the design to compile correctly.

Nearly all Vivado physical optimizations work on redacted netlists. The only exception is `-restruct_opt`, which restructures logic based on LUT INIT values. Disabling this optimization results in only 2.8% geometric f_{max} degradation across all benchmarks. Most importantly, all benchmarks produce bitstream-identical results after restoration, proving that the redaction process does not alter design functionality.

Design	Cells	Δf_{max}	Match
stereovision1	30404	-1.0%	✓
jpegencode	65751	-5.1%	✓
aes128	8577	-0.7%	✓
blob_merge	7858	-15.1%	✓
or1200	3741	-8.2%	✓
sha	2422	-1.9%	✓
basicsrsa	1306	1.6%	✓
diffeq1	1029	0.9%	✓
GeoMean		-2.8%	

Redaction results on selected benchmark designs